

# IS105 - ENTERPRISE ARCHITECTURE

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# ENTERPRISE ARCHITECTURE TOOLS (PART 2)



## **AT THE END OF THIS LESSON, YOU SHOULD BE ABLE TO:**

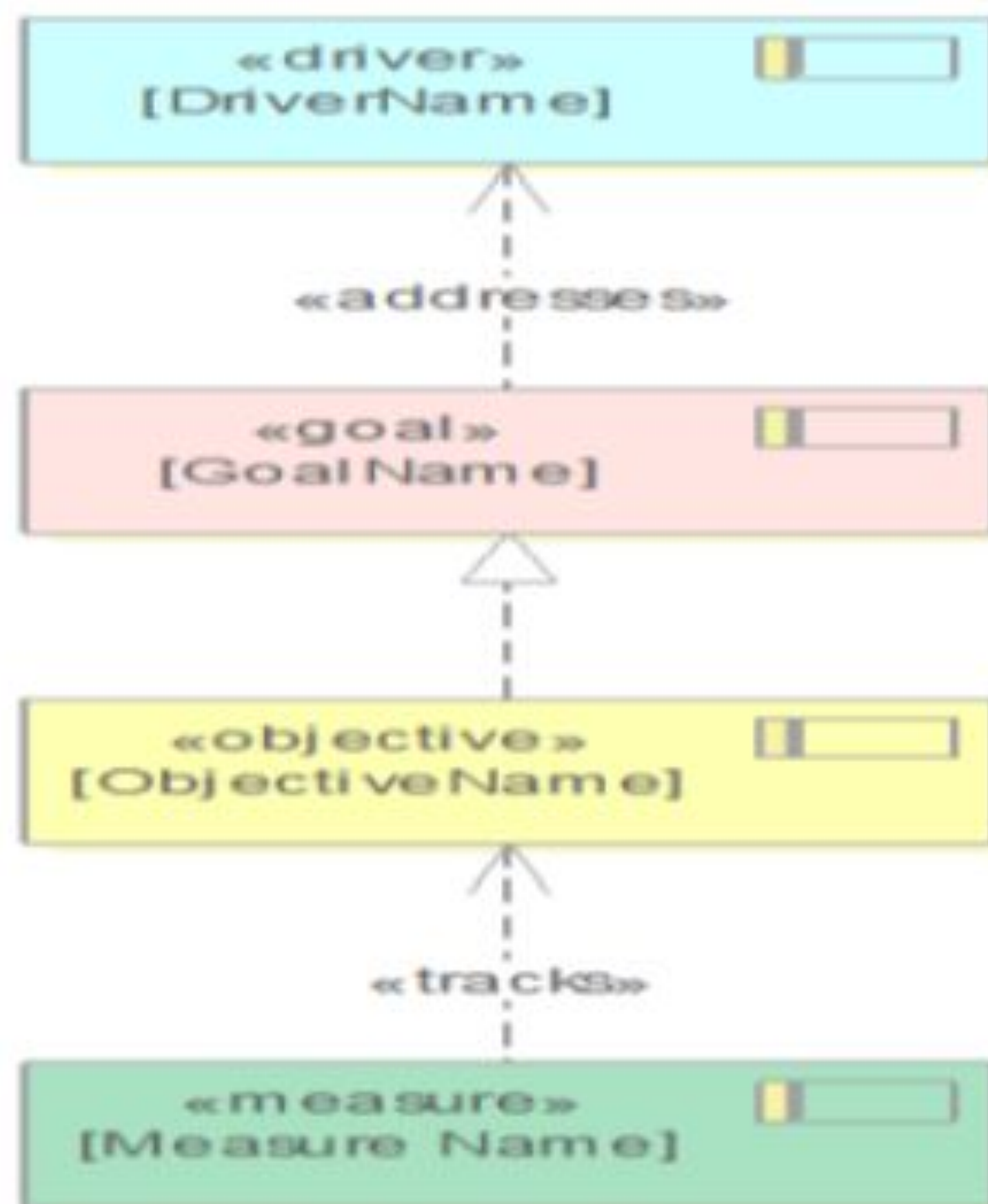
- Familiarize with the Enterprise Architect tool.
- Be conscious how important Enterprise Architect both as an architecture repository and a tool for managing architecture projects.
- Learn how to use Enterprise Architect tool for planning, developing and documenting

# PATTERNS



# PATTERN

- Patterns provide reusable design solutions for recurring architectural problems, acting as templates rather than finalized designs.
- Enterprise Architect enables publishing any diagram as a pattern, allowing consistent reuse of structures like goal-measurement hierarchies across projects.
- Originating from building architecture and adapted for software engineering, they standardize approaches across diverse contexts like business processes or technology frameworks.



# RELATIONSHIP MATRIX



# RELATIONSHIP MATRIX

- This matrix visualizes relationships between model elements (e.g., Stakeholders $\leftrightarrow$ Requirements) in a spreadsheet-like grid for immediate gap analysis.
- It identifies missing connections, such as unlinked requirements or unvalidated components, enhancing traceability during stakeholder workshops.
- Saved profiles enable rapid reuse for comparative analysis across projects or documentation exports to CSV formats.

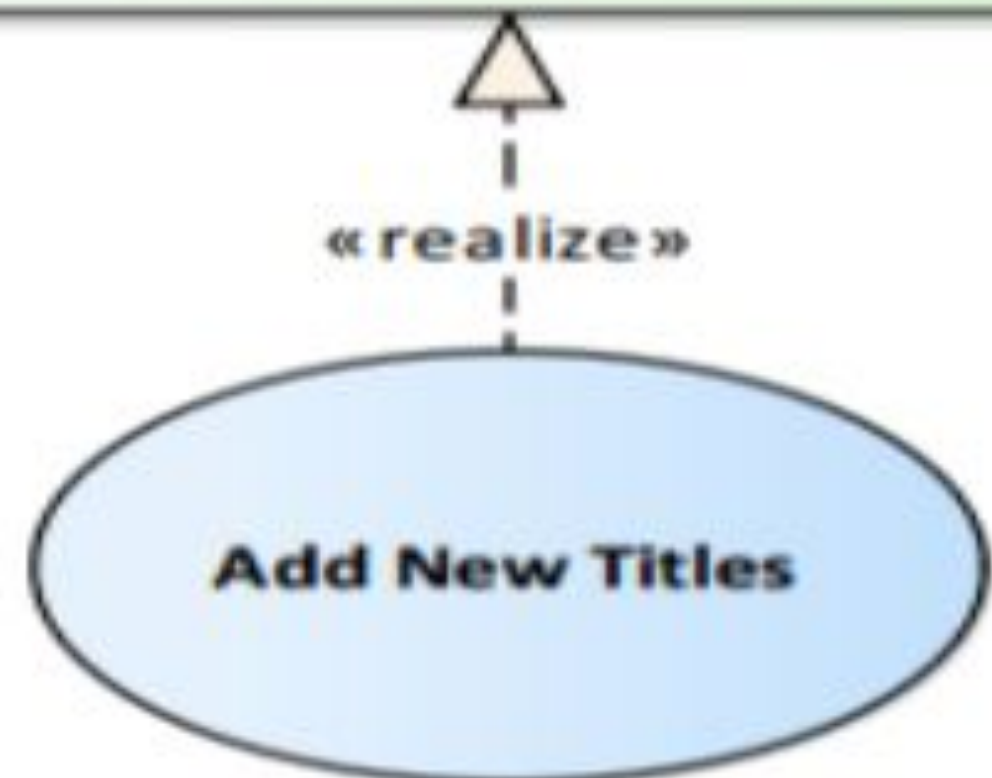
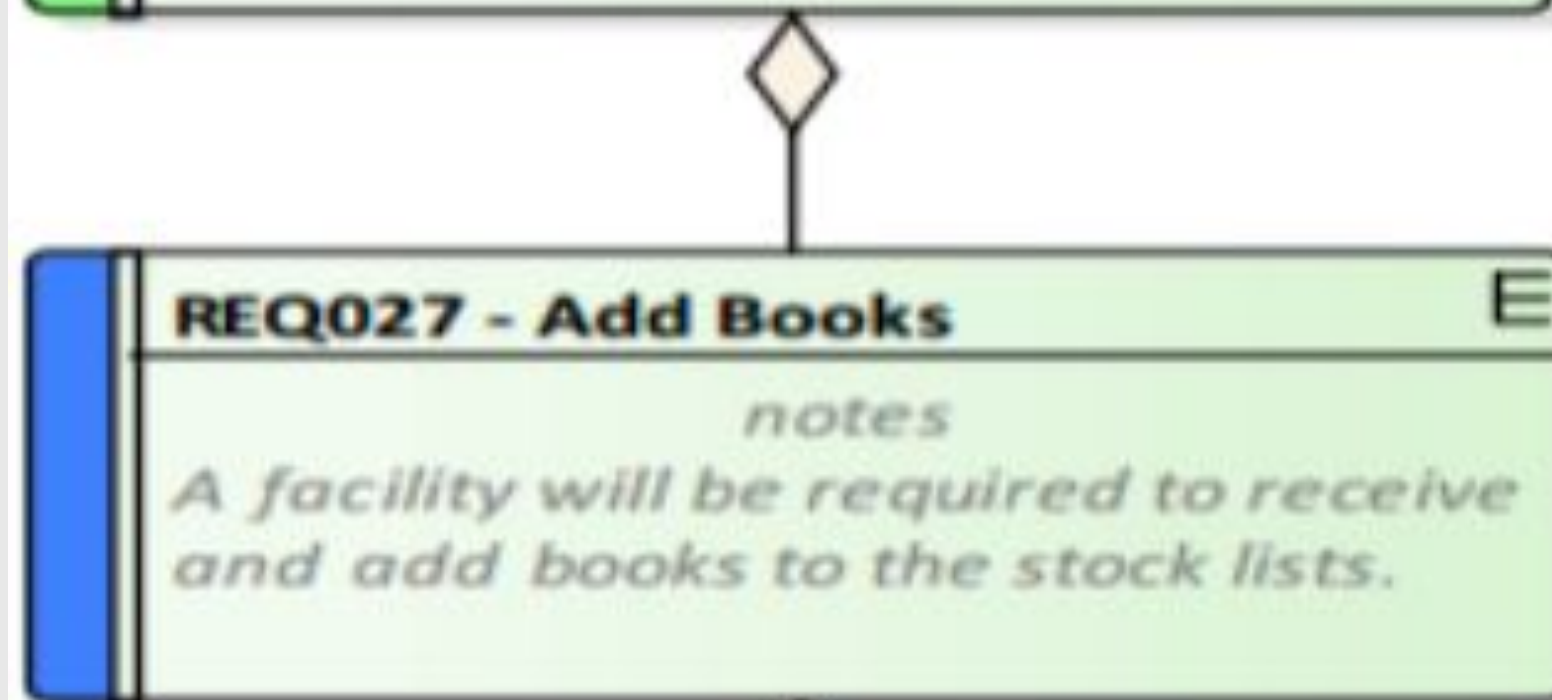
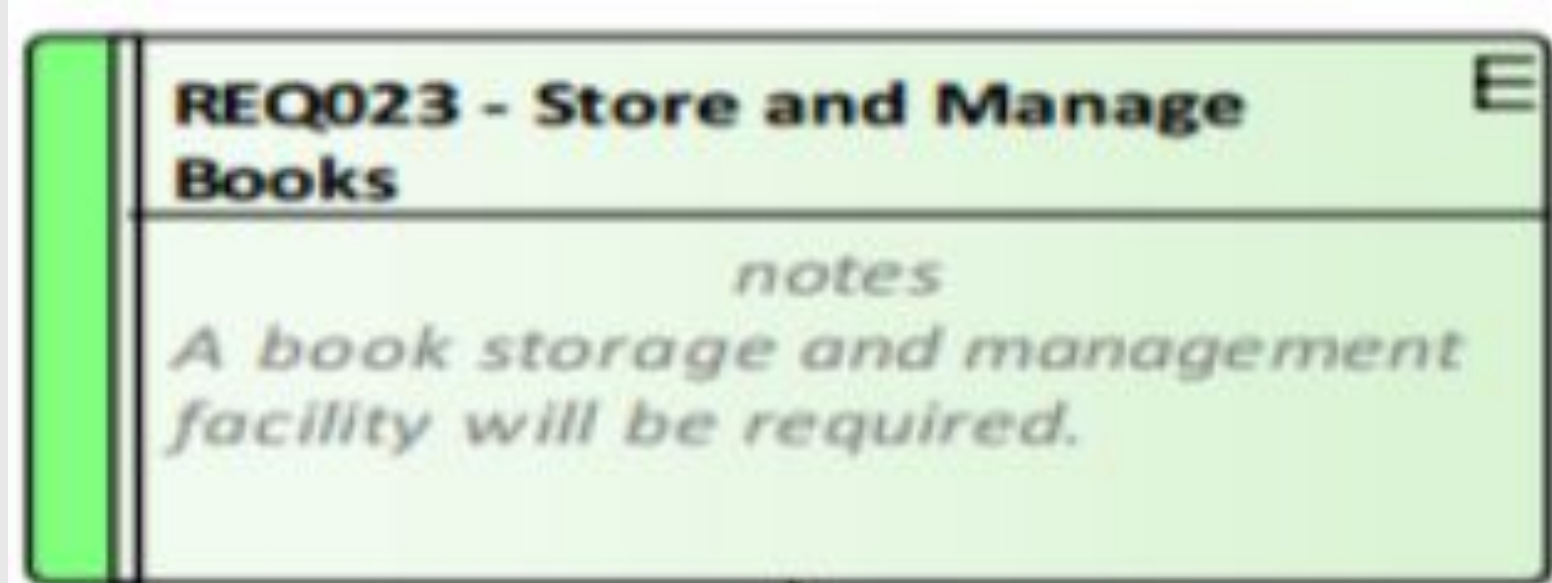
| <div> <div>Target</div> <div>+</div> </div> <div> <div>+</div> <div>Source</div> </div> | REQ011 - Manage User Accounts | REQ012 - Provide Online Sales | REQ013 - Manage Deliveries | REQ014 - ShoppingBasket | REQ015 - Process Credit Card Payment | REQ016 - Add Users | REQ017 - Remove User | REQ018 - Report on User Account | REQ019 - Manage Inventory | REQ020 - Receive Books | REQ021 - List Stock Levels | REQ022 - Order Books |
|-----------------------------------------------------------------------------------------|-------------------------------|-------------------------------|----------------------------|-------------------------|--------------------------------------|--------------------|----------------------|---------------------------------|---------------------------|------------------------|----------------------------|----------------------|
| Add New Titles                                                                          |                               |                               |                            |                         |                                      |                    |                      |                                 |                           |                        |                            |                      |
| Add To Shopping Basket                                                                  |                               |                               |                            | ↑                       |                                      |                    |                      |                                 |                           |                        |                            |                      |
| Close Account                                                                           |                               |                               |                            |                         |                                      |                    | ↑                    |                                 |                           |                        |                            |                      |
| Create Account                                                                          |                               |                               |                            |                         |                                      | ↑                  |                      |                                 |                           |                        |                            |                      |
| Create Orders                                                                           |                               |                               |                            |                         |                                      |                    |                      |                                 |                           |                        |                            | ↑                    |
| Delete User                                                                             |                               |                               |                            |                         |                                      |                    | ↑                    |                                 |                           |                        |                            |                      |

# REQUIREMENT S DIAGRAM



# REQUIREMENTS

- **DIAGRAM** Requirements Diagrams visually map hierarchical relationships between requirements and connected elements like business rules, use cases, and components.
- They transform text-based requirements into intuitive graphical networks, clarifying dependencies that inform solution design and validation.
- When integrated with the Specification Manager, they ensure consistency between visual models and textual requirement repositories.



*(from Manage Inventory)*

This diagram shows a Use Case that realizes a requirement. The realized requirement is part of a hierarchy of requirements expressed with the Aggregation relationship. The Use Case has an annotation under the element indicating its package location in the Project Browser.

# ROADMAP DIAGRAM

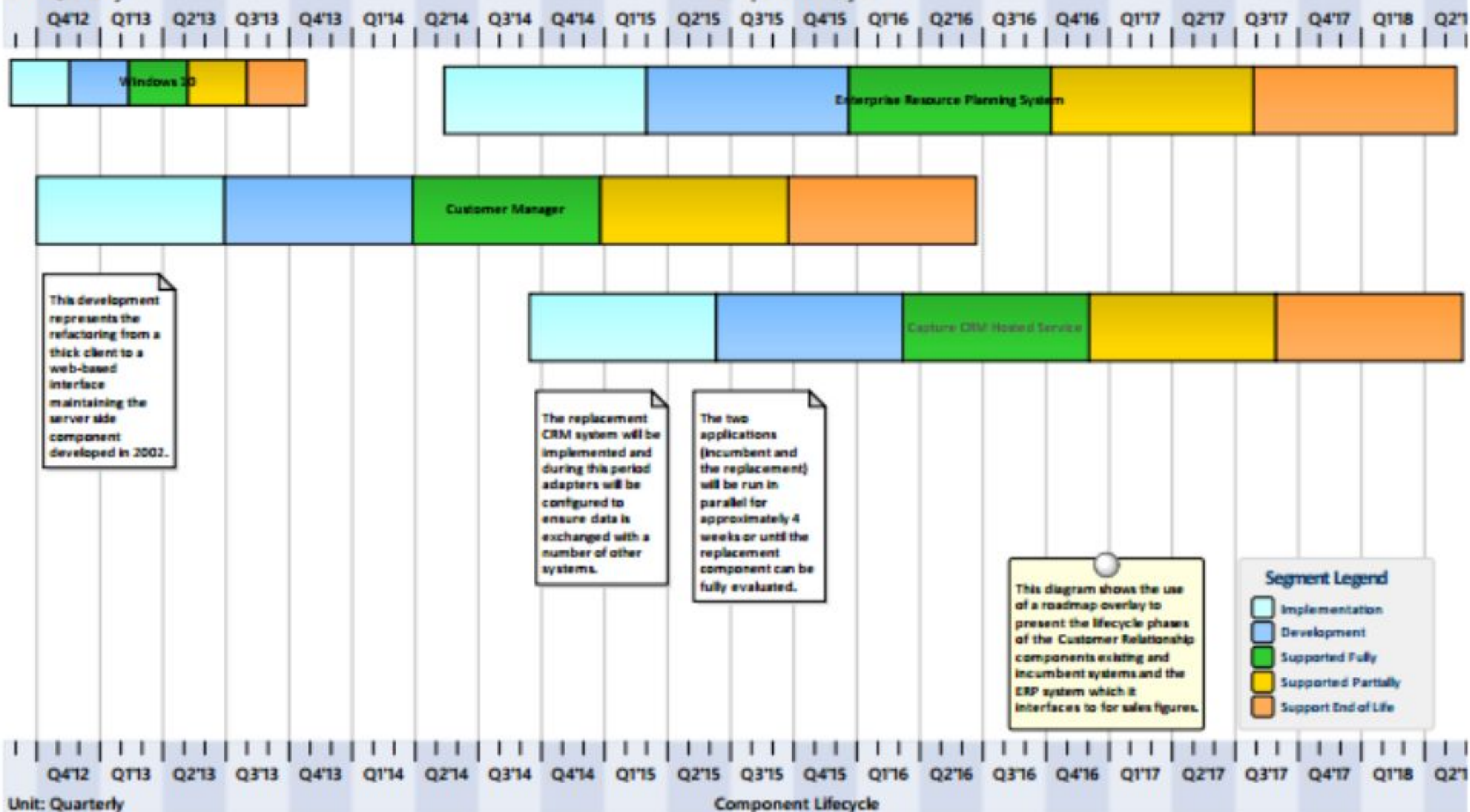


# ROADMAP DIAGRAM

- Roadmap Diagrams overlay timelines on any diagram to show element evolution through user-defined phases like current/future states.
- They model strategic transitions such as application lifecycles or capability maturities using configurable color-coded bands.
- Legends customize phase displays, enabling focused presentations by hiding irrelevant segments during stakeholder reviews.

Unit: Quarterly

Component Lifecycle



# SPECIFICATION MANAGER



# SPECIFICATION

- **MANAGER** As the central requirements hub, it provides a document-style interface for bulk-editing properties like status, priority, and descriptions.
- Changes automatically synchronize across all model views, maintaining consistency in diagrams, matrices, and documentation.
- Advanced features include hierarchical numbering, auto-naming, spell check, and import/export for end-to-end governance.

## **1 REQ019 - Manage Inventory**

The system **MUST** include a complete inventory management facility to store and track stock of books for the on-line bookstore.

### **1.1 REQ122 - Inventory Reports**

Inventory reports are required that detail the available stock for each item including back orders. Future stock level reports should be able to predict the quantity of stock at a specified future date.

### **1.2 REQ023 - Store and Manage Books**

A book storage and management facility will be required.

#### **1.2.1 REQ022 - Order Books**

A book order facility will be required to allow on-line ordering from major stockist's.

#### **1.2.2 REQ021 - List Stock Levels**

A facility will exist to list current stock levels and to manually update stock quantities if physical checking reveals inconsistencies.

# STRATEGY MAP



# STRATEGY MAP

- Strategy Maps align organizational goals across four perspectives: Financial, Customer, Internal Processes, and Learning/Growth.
- They communicate strategic priorities visually, linking high-level objectives to implementation elements like capabilities or projects.
- This tool fosters organizational alignment by clarifying how departmental initiatives contribute to overarching mission success.

Financial Perspective

Productivity Strategy

Growth Strategy

Long-Term Shareholder Value

Improve Cost Structure

Increase Asset Utilization

Expand Revenue Opportunities

Enhance Customer Value

Customer Perspective

Customer Value Proposition

Price

Consistent Quality

High Availability

Titles Selection

Service Excellence

Partnership

Brand

Product or Service Attributes

Relationship

Image

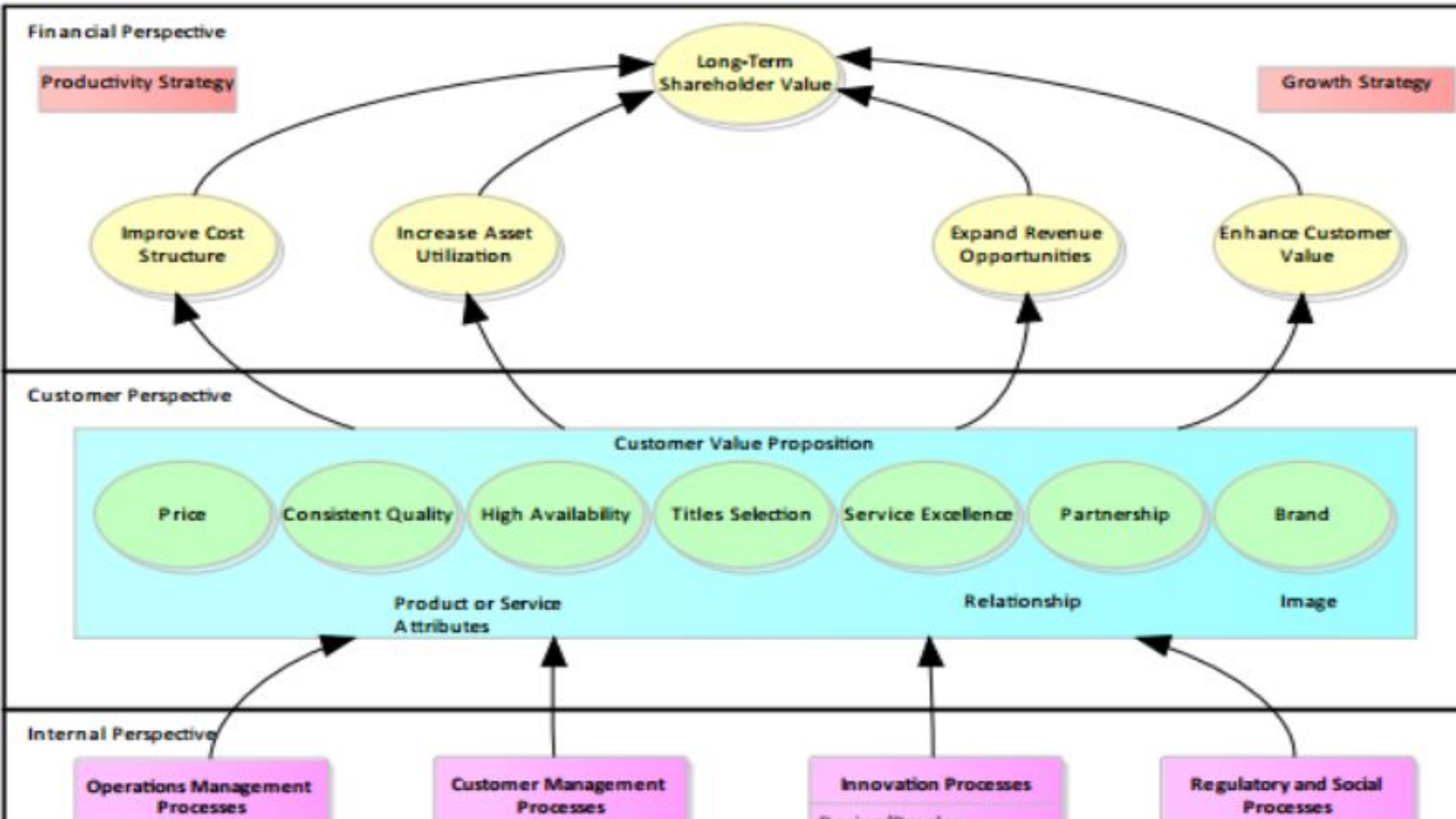
Internal Perspective

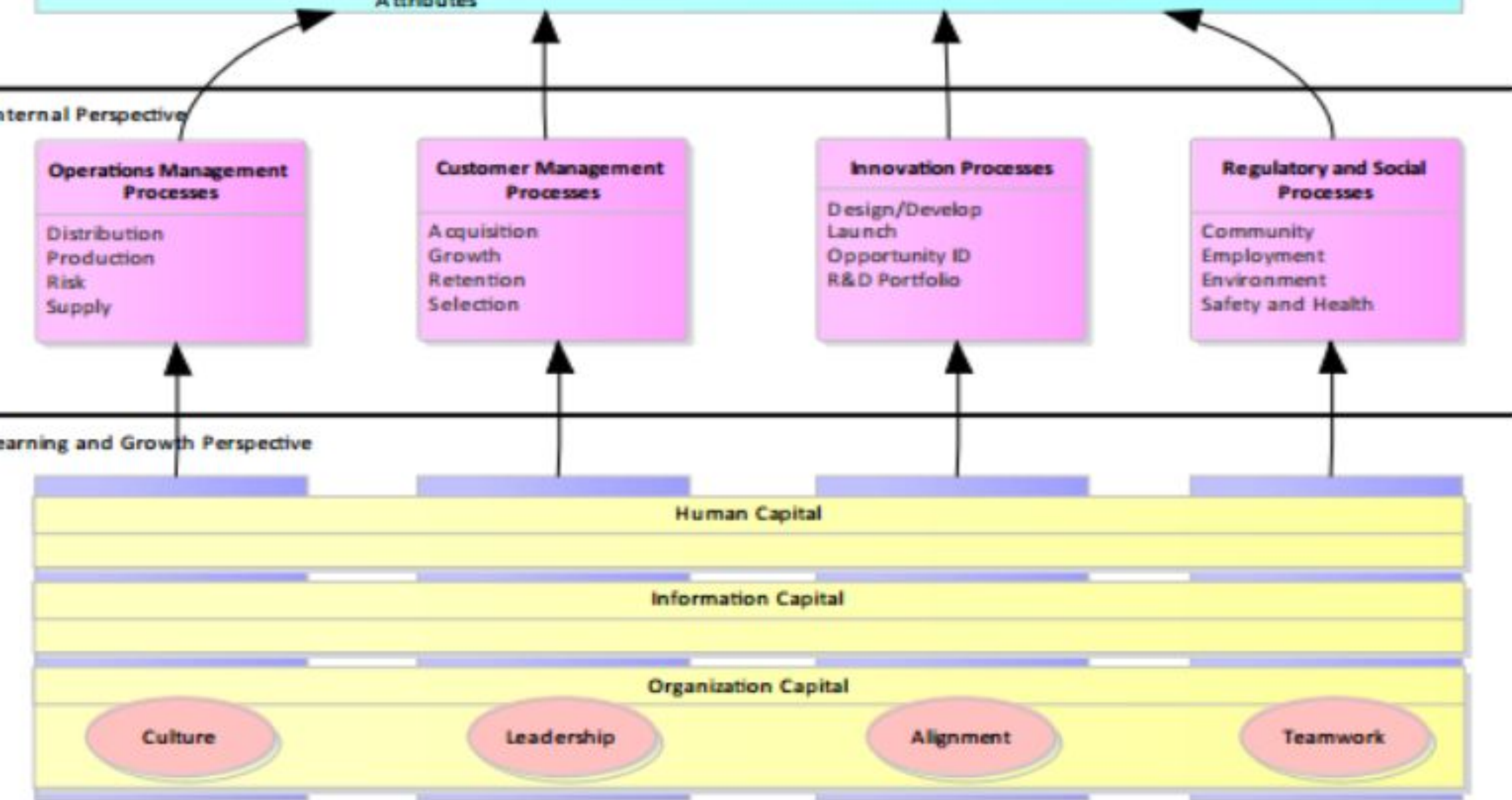
Operations Management Processes

Customer Management Processes

Innovation Processes

Regulatory and Social Processes





# TEAM REVIEWS

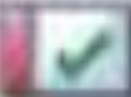


# TEAM REVIEWS

- Team Reviews facilitate structured feedback from stakeholders, developers, and architects during project milestones or model validations.
- They support diverse formats including walkthroughs, formal assessments, and ad-hoc collaboration to refine architectural outputs.
- Centralized comment tracking ensures actionable insights are captured, improving model accuracy and business-IT alignment.



-   **Deployment Issues**
-   **Requirement Issues**
  -   **Shopping Cart facility**
    -   Shopping Cart Button
    -   Model Links

 REQ014 - Shopping Basket

 **Use Case Model**

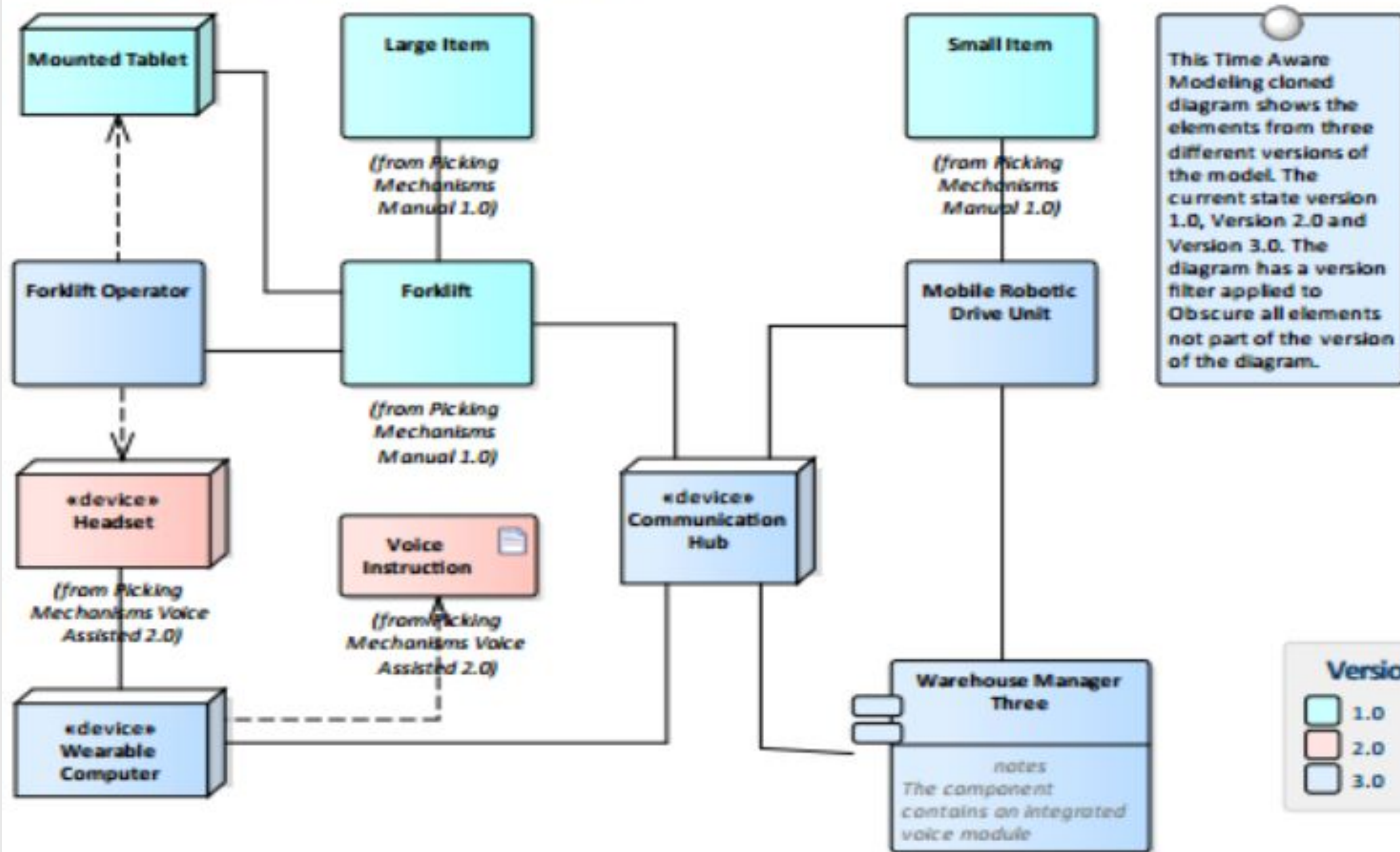
# TIME AWARE MODELING



# TIME AWARE

- **MODELING** This feature clones "As-Is" elements/diagrams into versioned "To-Be" states for temporal analysis of business or technology evolution.
- It enables "what-if" scenario planning by maintaining baseline models while simulating multiple future options for comparison.
- Filters isolate version-specific elements in diagrams, while traceability tools visualize cross-version dependencies.

# Integrated Mobile Robotic Drive Units and Voice Assisted Forklifts Version 3.0 Q4 2019

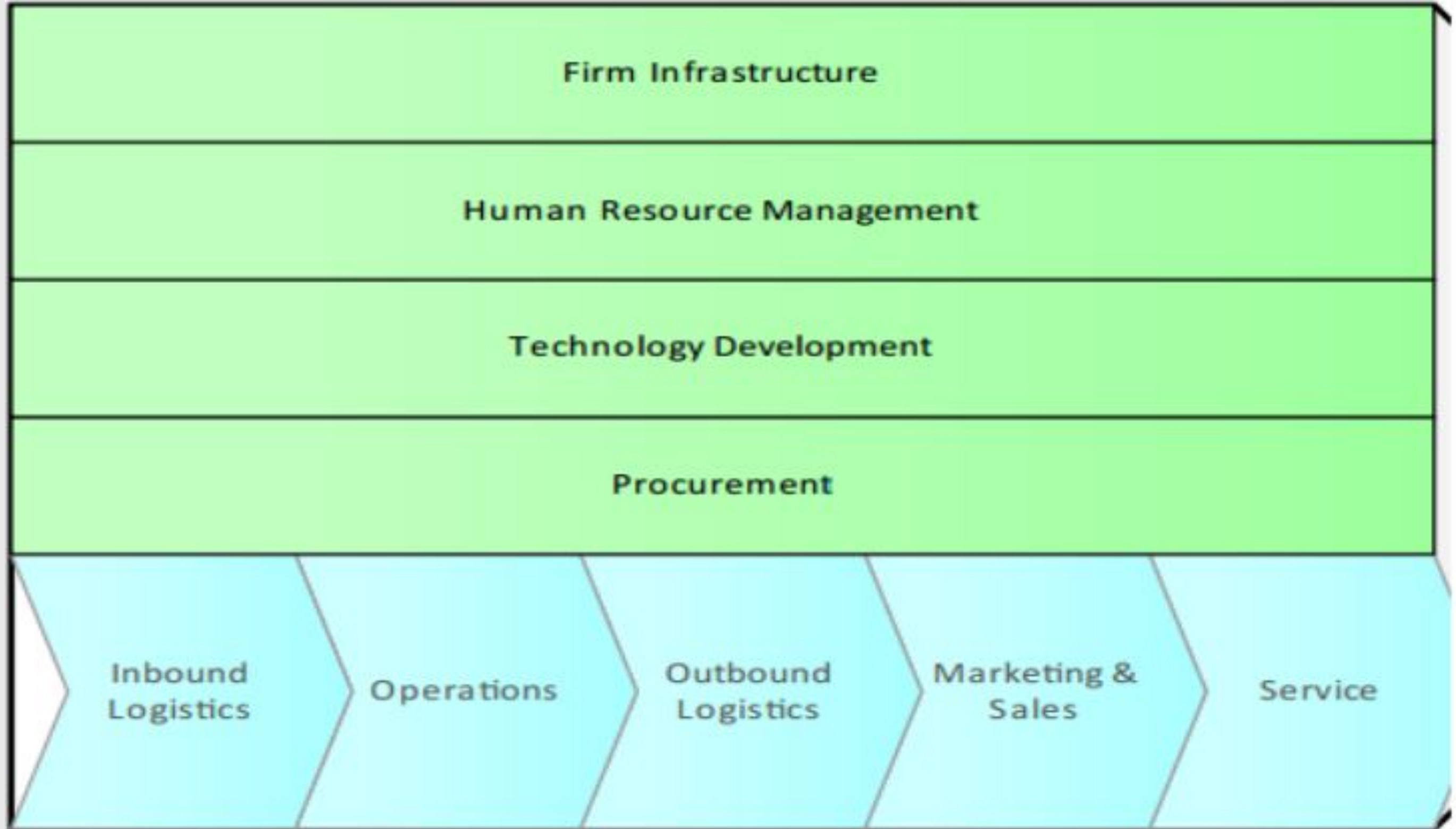


# VALUE CHAIN



# VALUE CHAIN

- Value Chains model primary activities (e.g., Operations, Marketing) and support activities (e.g., HR, Procurement) that generate customer value.
- They analyze cost/resource allocation across the chain, identifying optimization opportunities for competitive advantage.
- Linked elements attach benchmarks or KPIs to activities, enabling quantitative performance evaluation and strategic refinement.



### Legend

- ☐ Gross Sales
- ☐ Support Activities
- ☐ Primary Activities

THE END